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Therefore $\Delta U_c = q_1 + q_2$

$= k.\Delta T + m_w C_w \Delta T$

$= (k+m_w C_w)\Delta T$

Calorimeter constant can be determined by burning a known mass of standard sample (benzoic acid) for which the heat of combustion is known (-3227 kJmol^{-1})

The enthalpy of combustion at constant pressure of the substance is calculated from the equation (7.17)

$\Delta H_C^0{}_{(pressure)} = \Delta U_C^0{}_{(Vol)} + \Delta n_g RT$

Applications of bomb calorimeter:

1. Bomb calorimeter is used to determine the amount of heat released in combustion reaction.

2. It is used to determine the calorific value of food.

3. Bomb calorimeter is used in many industries such as metabolic study, food processing, explosive testing etc.

(b) ΔH Measurements

Heat change at constant pressure (at atmospheric pressure) can be measured using a coffee cup calorimeter. A schematic representation of a coffee cup calorimeter is given in Figure 7.7. Instead of bomb, a styrofoam cup is used in this calorimeter. It acts as good adiabatic wall and doesn't allow transfer of heat produced during the reaction to its surrounding. This entire heat energy is absorbed by the water inside the cup. This method can be used for the reactions where there is no appreciable change in volume. The change in the temperature of water is measured and used to calculate the amount of heat that has been absorbed or evolved in the reaction using the following expression.

$$q = m_w C_w \Delta T$$

where m_w is the molar mass of water and C_w is the molar heat capacity of water (4184 kJ K^{-1} mol^{-1})

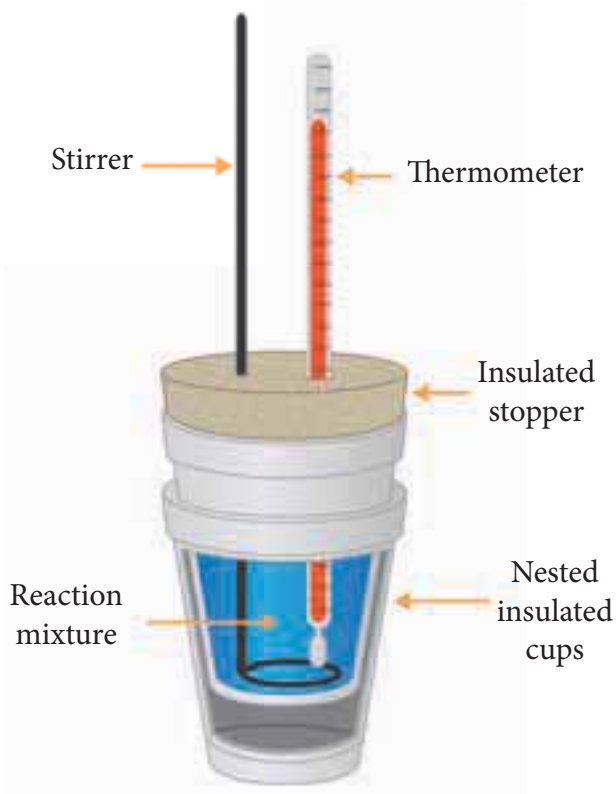


Figure 7.7 Coffee cup Calorimeter

Problem 7. 4

Calculate the enthalpy of combustion of ethylene at 300 K at constant pressure, if its heat of combustion at constant volume (ΔU) is −1406 kJ .

The complete ethylene combustion reaction can be written as,